DB Logging for Auditing

Audit log

Record every important action.

Who accessed what from where and when?

Written to stable storage.

Why?

Compliance

- FERPA, HIPAA, GDPR, …

Security

Audit log ≠ transaction log

- Different purposes
- Audit log durable, but not necessarily atomic.

How?

Explicitly in DB code

- Triggers
- API functions

- Audit info in tables or separate file

Implicitly by DBMS (or extensions)

- Need to specify what to log, but system handles details.

PostgreSQL: Not built-in. Extensions available, or do it yourself.

Example trigger

```
CREATE OR REPLACE FUNCTION EmpTrig()
RETURNS TRIGGER
AS $$
    BEGIN
        NEW.emp_last_update := current_timestamp;
        NEW.emp_updatedby := current_user;
        RETURN NEW;
    END $$
LANGUAGE plpgsql;

CREATE TRIGGER emp_trig BEFORE INSERT OR UPDATE ON Employee
    FOR EACH ROW EXECUTE PROCEDURE EmpTrig();
```